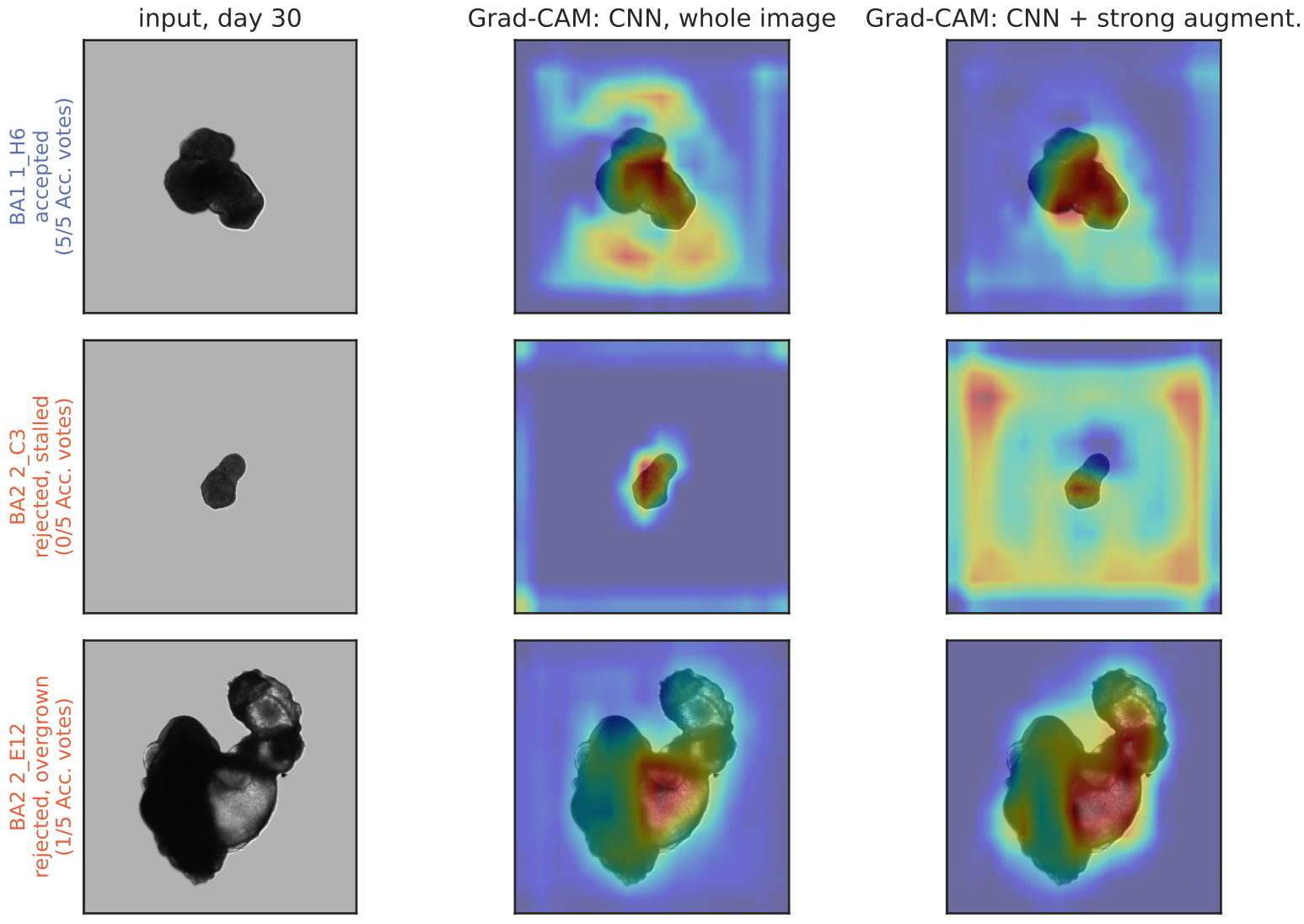
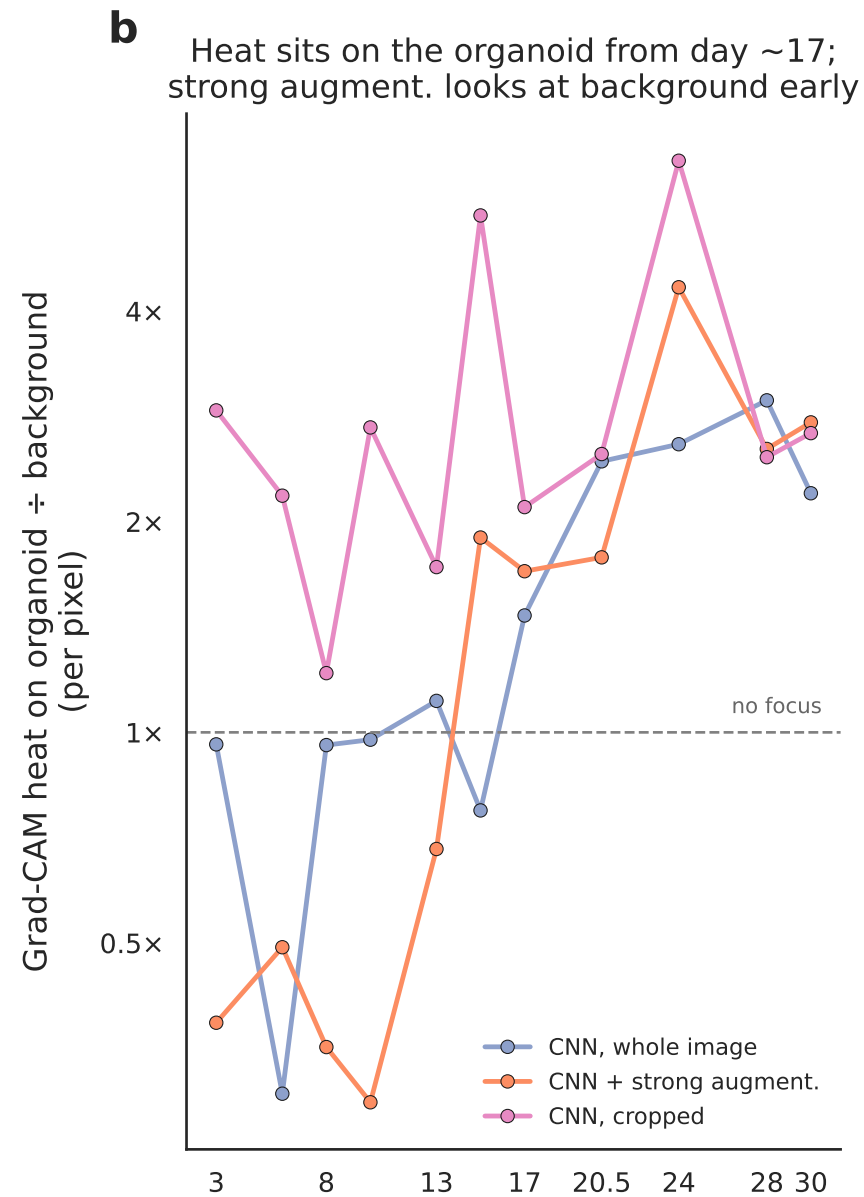
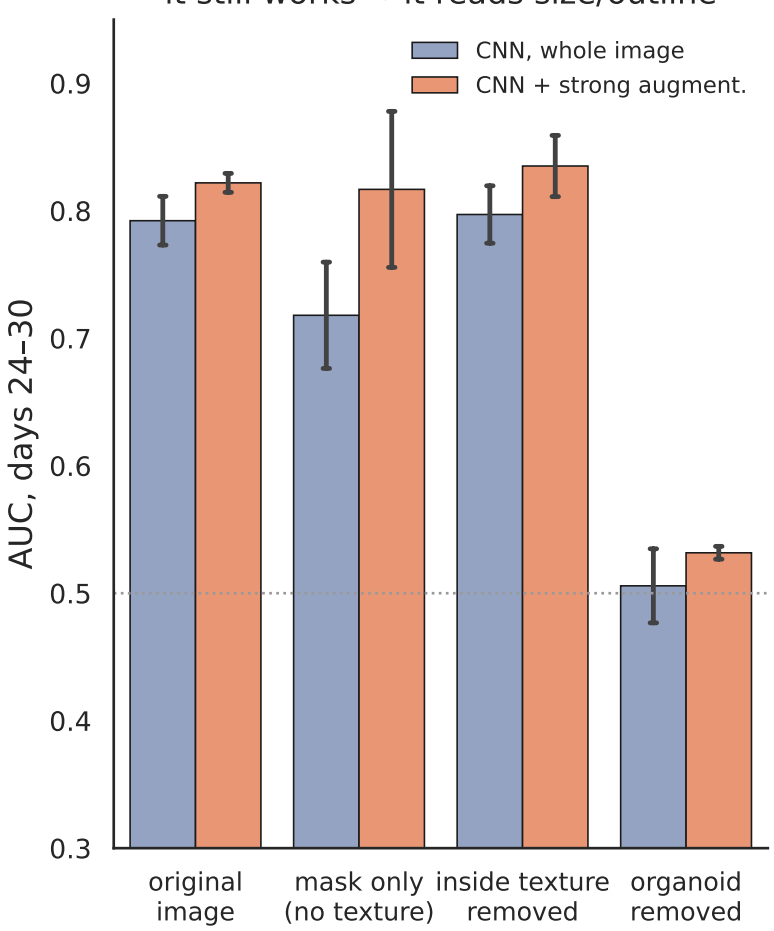
a


input, day 30
Grad-CAM: CNN, whole image
Grad-CAM: CNN + strong augment.
BA1 1_H6
accepted
(5/5 Acc. votes)
BA2 2_C3
rejected, stalled
(0/5 Acc. votes)
BA2 2_E12
rejected, overgrown
(1/5 Acc. votes)

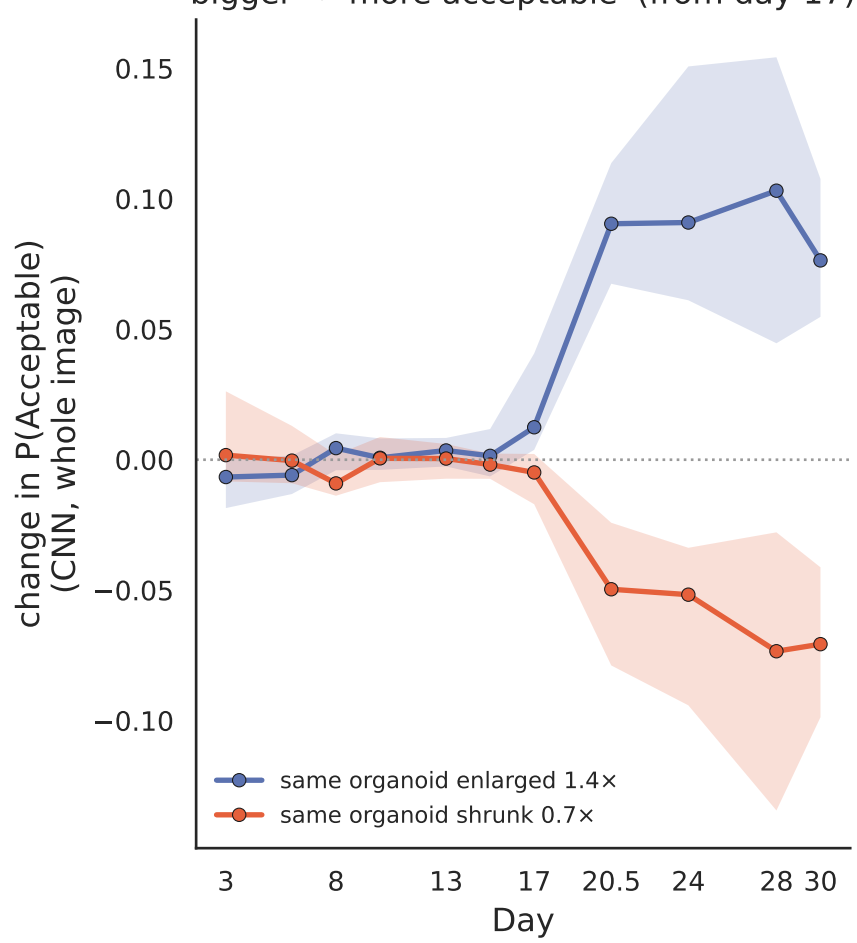
b


Heat sits on the organoid from day ~17; strong augment. looks at background early
Grad-CAM heat on organoid ÷ background (per pixel)
4×
2×
1×
0.5×
no focus
CNN, whole image
CNN + strong augment.
CNN, cropped
3
8
13
17
20.5
24
28
30
Day

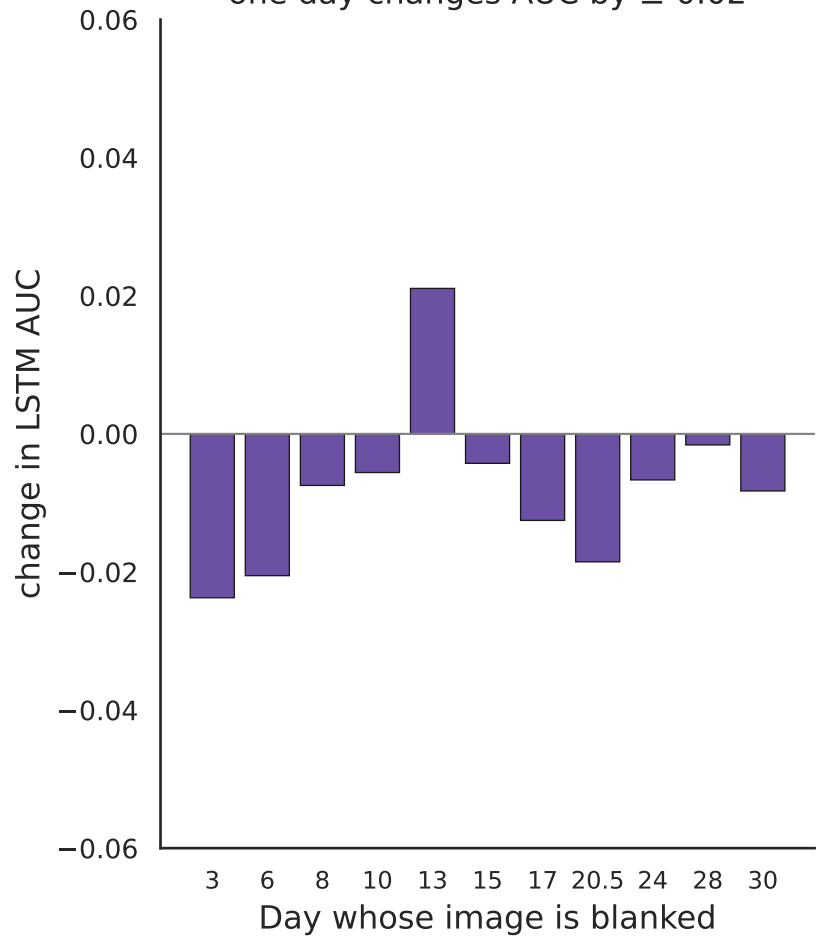
c


Show the trained CNN only the mask: it still works → it reads size/outline
CNN, whole image
CNN + strong augment.
AUC, days 24–30
0.9
0.8
0.7
0.6
0.5
0.4
0.3
original image
mask only (no texture)
inside texture removed
organoid removed

d


Only resize the organoid: bigger → 'more acceptable' (from day 17)
change in P(Acceptable) (CNN, whole image)
0.15
0.10
0.05
0.00
−0.05
−0.10
same organoid enlarged 1.4×
same organoid shrunk 0.7×
3
8
13
17
20.5
24
28
30
Day

e


LSTM (averaged over frames): blanking any one day changes AUC by ≤ 0.02
change in LSTM AUC
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
3
6
8
10
13
15
17
20.5
24
28
30
Day whose image is blanked